

Formulaire opérateurs différentiels

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1 Coordonnées cartésiennes

$$\overrightarrow{\text{OM}} = x \vec{e}_x + y \vec{e}_y + z \vec{e}_z$$

$$d\overrightarrow{\text{OM}} = dx \vec{e}_x + dy \vec{e}_y + dz \vec{e}_z$$

$$dV = dx \cdot dy \cdot dz$$

$$\nabla = \vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z}$$

Soit un champ scalaire $f(x, y, z)$.

$$\overrightarrow{\text{grad}} f = \nabla f = \frac{\partial f}{\partial x} \vec{e}_x + \frac{\partial f}{\partial y} \vec{e}_y + \frac{\partial f}{\partial z} \vec{e}_z$$

$$\Delta f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Soit un champ vectoriel $\vec{U}(x, y, z) = U_x \vec{e}_x + U_y \vec{e}_y + U_z \vec{e}_z$.

$$\overrightarrow{\text{grad}} \vec{U} = \nabla \vec{U} = \begin{pmatrix} \frac{\partial U_x}{\partial x} & \frac{\partial U_x}{\partial y} & \frac{\partial U_x}{\partial z} \\ \frac{\partial U_y}{\partial x} & \frac{\partial U_y}{\partial y} & \frac{\partial U_y}{\partial z} \\ \frac{\partial U_z}{\partial x} & \frac{\partial U_z}{\partial y} & \frac{\partial U_z}{\partial z} \end{pmatrix}_{(\vec{e}_x, \vec{e}_y, \vec{e}_z)}$$

$$\text{div} \vec{U} = \nabla \cdot \vec{U} = \frac{\partial U_x}{\partial x} + \frac{\partial U_y}{\partial y} + \frac{\partial U_z}{\partial z}$$

$$\overrightarrow{\text{rot}} \vec{U} = \nabla \wedge \vec{U} = \left(\frac{\partial U_z}{\partial y} - \frac{\partial U_y}{\partial z} \right) \vec{e}_x + \left(\frac{\partial U_x}{\partial z} - \frac{\partial U_z}{\partial x} \right) \vec{e}_y + \left(\frac{\partial U_y}{\partial x} - \frac{\partial U_x}{\partial y} \right) \vec{e}_z$$

$$\Delta \vec{U} = \nabla^2 \vec{U} = \Delta U_x \vec{e}_x + \Delta U_y \vec{e}_y + \Delta U_z \vec{e}_z$$

Soit un champ tensoriel d'ordre 2 $\underline{\underline{A}}(x, y, z) = \begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{yx} & A_{yy} & A_{yz} \\ A_{zx} & A_{zy} & A_{zz} \end{pmatrix}_{(\vec{e}_x, \vec{e}_y, \vec{e}_z)}$

$$\begin{aligned} \overrightarrow{\text{div}} \underline{\underline{A}} &= \nabla \cdot \underline{\underline{A}} = \left(\frac{\partial A_{xx}}{\partial x} + \frac{\partial A_{xy}}{\partial y} + \frac{\partial A_{xz}}{\partial z} \right) \vec{e}_x \\ &+ \left(\frac{\partial A_{yx}}{\partial x} + \frac{\partial A_{yy}}{\partial y} + \frac{\partial A_{yz}}{\partial z} \right) \vec{e}_y \\ &+ \left(\frac{\partial A_{zx}}{\partial x} + \frac{\partial A_{zy}}{\partial y} + \frac{\partial A_{zz}}{\partial z} \right) \vec{e}_z \end{aligned}$$

2 Coordonnées cylindriques

$$\overrightarrow{OM} = r \vec{e}_r + z \vec{e}_z$$

$$d\overrightarrow{OM} = dr \vec{e}_r + r d\theta \vec{e}_\theta + dz \vec{e}_z$$

$$d\vec{S}_b = dS_b \vec{e}_r = dr \cdot r d\theta \vec{e}_r \quad d\vec{S}_l = dS_l \vec{e}_z = r d\theta \cdot dz \vec{e}_z$$

$$dV = dr \cdot r d\theta \cdot dz$$

$$\nabla = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{e}_z \frac{\partial}{\partial z}$$

Soit un champ scalaire $f(r, \theta, z)$.

$$\overrightarrow{\text{grad}} f = \nabla f = \frac{\partial f}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \vec{e}_\theta + \frac{\partial f}{\partial z} \vec{e}_z$$

$$\Delta f = \nabla^2 f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}$$

Soit un champ vectoriel $\vec{U}(r, \theta, z) = U_r \vec{e}_r + U_\theta \vec{e}_\theta + U_z \vec{e}_z$.

$$\overrightarrow{\text{grad}} \vec{U} = \nabla \vec{U} = \begin{pmatrix} \frac{\partial U_r}{\partial r} & \frac{1}{r} \left(\frac{\partial U_r}{\partial \theta} - U_\theta \right) & \frac{\partial U_r}{\partial z} \\ \frac{\partial U_\theta}{\partial r} & \frac{1}{r} \left(\frac{\partial U_\theta}{\partial \theta} + U_r \right) & \frac{\partial U_\theta}{\partial z} \\ \frac{\partial U_z}{\partial r} & \frac{1}{r} \frac{\partial U_z}{\partial \theta} & \frac{\partial U_z}{\partial z} \end{pmatrix}_{(\vec{e}_r, \vec{e}_\theta, \vec{e}_z)}$$

$$\text{div} \vec{U} = \nabla \cdot \vec{U} = \frac{1}{r} \frac{\partial}{\partial r} (r U_r) + \frac{1}{r} \frac{\partial U_\theta}{\partial \theta} + \frac{\partial U_z}{\partial z}$$

$$\overrightarrow{\text{rot}} \vec{U} = \nabla \wedge \vec{U} = \left(\frac{1}{r} \frac{\partial U_z}{\partial \theta} - \frac{\partial U_\theta}{\partial z} \right) \vec{e}_r + \left(\frac{\partial U_r}{\partial z} - \frac{\partial U_z}{\partial r} \right) \vec{e}_\theta + \left(\frac{1}{r} \frac{\partial (r U_\theta)}{\partial r} - \frac{1}{r} \frac{\partial U_r}{\partial \theta} \right) \vec{e}_z$$

$$\Delta \vec{U} = \nabla^2 \vec{U} = \left(\Delta U_r - \frac{2}{r^2} \frac{\partial U_\theta}{\partial \theta} - \frac{U_r}{r^2} \right) \vec{e}_r + \left(\Delta U_\theta + \frac{2}{r^2} \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r^2} \right) \vec{e}_\theta + (\Delta U_z) \vec{e}_z$$

Soit un champ tensoriel d'ordre 2 $\underline{\underline{A}}(r, \theta, z) = \begin{pmatrix} A_{rr} & A_{r\theta} & A_{rz} \\ A_{\theta r} & A_{\theta\theta} & A_{\theta z} \\ A_{zr} & A_{z\theta} & A_{zz} \end{pmatrix}_{(\vec{e}_r, \vec{e}_\theta, \vec{e}_z)}$

$$\begin{aligned} \overrightarrow{\text{div}} \underline{\underline{A}} = \nabla \cdot \underline{\underline{A}} = & \left(\frac{\partial A_{rr}}{\partial r} + \frac{1}{r} \frac{\partial A_{\theta r}}{\partial \theta} + \frac{\partial A_{zr}}{\partial z} + \frac{A_{rr} - A_{\theta\theta}}{r} \right) \vec{e}_r \\ & + \left(\frac{\partial A_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{\partial A_{z\theta}}{\partial z} + \frac{A_{r\theta} + A_{\theta r}}{r} \right) \vec{e}_\theta \\ & + \left(\frac{\partial A_{rz}}{\partial r} + \frac{1}{r} \frac{\partial A_{\theta z}}{\partial \theta} + \frac{\partial A_{zz}}{\partial z} + \frac{A_{rz}}{r} \right) \vec{e}_z \end{aligned}$$

Soit un champ tensoriel d'ordre 2 symétrique $\underline{\underline{T}}(r, \theta, z) = \begin{pmatrix} T_{rr} & T_{r\theta} & T_{rz} \\ & T_{\theta\theta} & T_{\theta z} \\ & & T_{zz} \end{pmatrix}_{(\vec{e}_r, \vec{e}_\theta, \vec{e}_z)}$

$$\begin{aligned}\overrightarrow{\text{div}} \underline{T} = \nabla \cdot \underline{T} &= \left(\frac{\partial T_{rr}}{\partial r} + \frac{1}{r} \frac{\partial T_{r\theta}}{\partial \theta} + \frac{\partial T_{rz}}{\partial z} + \frac{T_{rr} - T_{\theta\theta}}{r} \right) \vec{e}_r \\ &+ \left(\frac{\partial T_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial T_{\theta\theta}}{\partial \theta} + \frac{\partial T_{\theta z}}{\partial z} + 2 \frac{T_{r\theta}}{r} \right) \vec{e}_\theta \\ &+ \left(\frac{\partial T_{rz}}{\partial r} + \frac{1}{r} \frac{\partial T_{\theta z}}{\partial \theta} + \frac{\partial T_{zz}}{\partial z} + \frac{T_{rz}}{r} \right) \vec{e}_z\end{aligned}$$

3 Coordonnées sphériques

$$\overrightarrow{\text{OM}} = r \vec{e}_r$$

$$d\overrightarrow{\text{OM}} = dr \vec{e}_r + r d\theta \vec{e}_\theta + r \sin \theta d\varphi \vec{e}_\varphi$$

$$d\vec{S} = dS \vec{e}_r = r d\theta \cdot r \sin \theta d\varphi \vec{e}_r$$

$$dV = dr \cdot r d\theta \cdot r \sin \theta d\varphi$$

$$\nabla = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi}$$

Soit un champ scalaire $f(r, \theta, \varphi)$.

$$\overrightarrow{\text{grad}} f = \nabla f = \frac{\partial f}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \vec{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \vec{e}_\varphi$$

$$\Delta f = \nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{1}{r^2 \tan \theta} \frac{\partial f}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2}$$

Soit un champ vectoriel $\vec{U}(r, \theta, \varphi) = U_r \vec{e}_r + U_\theta \vec{e}_\theta + U_\varphi \vec{e}_\varphi$.

$$\overrightarrow{\text{grad}} \vec{U} = \nabla \vec{U} = \begin{pmatrix} \frac{\partial U_r}{\partial r} & \frac{1}{r} \left(\frac{\partial U_r}{\partial \theta} - U_\theta \right) & \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial U_r}{\partial \varphi} - U_\varphi \right) \\ \frac{\partial U_\theta}{\partial r} & \frac{1}{r} \left(\frac{\partial U_\theta}{\partial \theta} + U_r \right) & \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial U_\theta}{\partial \varphi} - \frac{U_\varphi}{\tan \theta} \right) \\ \frac{\partial U_\varphi}{\partial r} & \frac{1}{r} \frac{\partial U_\varphi}{\partial \theta} & \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial U_\varphi}{\partial \varphi} + \frac{U_\theta}{\tan \theta} + U_r \right) \end{pmatrix}_{(\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi)}$$

$$\text{div} \vec{U} = \nabla \cdot \vec{U} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 U_r) + \frac{1}{r} \frac{\partial U_\theta}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial U_\varphi}{\partial \varphi} + \frac{U_\theta}{r \tan \theta}$$

$$\begin{aligned}\overrightarrow{\text{rot}} \vec{U} = \nabla \wedge \vec{U} &= \left(\frac{1}{r} \frac{\partial U_\varphi}{\partial \theta} - \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial U_\theta}{\partial \varphi} - \frac{U_\varphi}{\tan \theta} \right) \right) \vec{e}_r \\ &+ \left(\frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial U_r}{\partial \varphi} - U_\varphi \right) - \frac{\partial U_\varphi}{\partial r} \right) \vec{e}_\theta \\ &+ \left(\frac{\partial U_\theta}{\partial r} - \frac{1}{r} \left(\frac{\partial U_r}{\partial \theta} - U_\theta \right) \right) \vec{e}_\varphi\end{aligned}$$

$$\begin{aligned}\Delta \vec{U} = \nabla^2 \vec{U} = & \left(\Delta U_r - 2 \frac{U_r}{r^2} - \frac{2U_\theta}{r^2 \tan \theta} - \frac{2}{r^2} \frac{\partial U_\theta}{\partial \theta} - \frac{2}{r^2 \sin \theta} \frac{\partial U_\varphi}{\partial \varphi} \right) \vec{e}_r \\ & + \left(\Delta U_\theta + \frac{2}{r^2} \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r^2 \sin^2 \theta} - \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial U_\varphi}{\partial \varphi} \right) \vec{e}_\theta \\ & + \left(\Delta U_\varphi + \frac{2}{r^2 \sin^2 \theta} \frac{\partial U_r}{\partial \varphi} + \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial U_\theta}{\partial \varphi} - \frac{U_\varphi}{r^2 \sin^2 \theta} \right) \vec{e}_\varphi\end{aligned}$$

Soit un champ tensoriel d'ordre 2 $\underline{\underline{A}}(r, \theta, \varphi) = \begin{pmatrix} A_{rr} & A_{r\theta} & A_{r\varphi} \\ A_{\theta r} & A_{\theta\theta} & A_{\theta\varphi} \\ A_{\varphi r} & A_{\varphi\theta} & A_{\varphi\varphi} \end{pmatrix}_{(\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi)}$

$$\begin{aligned}\vec{\text{div}} \underline{\underline{A}} = \nabla \cdot \underline{\underline{A}} = & \left(\frac{\partial A_{rr}}{\partial r} + \frac{1}{r} \frac{\partial A_{\theta r}}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial A_{\varphi r}}{\partial \varphi} + \frac{2A_{rr} - A_{\theta\theta} - A_{\varphi\varphi} + A_{\theta r} \cot \theta}{r} \right) \vec{e}_r \\ & + \left(\frac{\partial A_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial A_{\theta\theta}}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial A_{\varphi\theta}}{\partial \varphi} + \frac{2A_{r\theta} + A_{\theta r} + (A_{\theta\theta} - A_{\varphi\varphi}) \cot \theta}{r} \right) \vec{e}_\theta \\ & + \left(\frac{\partial A_{r\varphi}}{\partial r} + \frac{1}{r} \frac{\partial A_{\theta\varphi}}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial A_{\varphi\varphi}}{\partial \varphi} + \frac{2A_{r\varphi} + A_{\varphi r} + (A_{\theta\varphi} + A_{\varphi\theta}) \cot \theta}{r} \right) \vec{e}_\varphi\end{aligned}$$

Soit un champ tensoriel d'ordre 2 symétrique $\underline{\underline{T}}(r, \theta, \varphi) = \begin{pmatrix} T_{rr} & T_{r\theta} & T_{r\varphi} \\ & T_{\theta\theta} & T_{\theta\varphi} \\ & & T_{\varphi\varphi} \end{pmatrix}_{(\vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi)}$

$$\begin{aligned}\vec{\text{div}} \underline{\underline{T}} = \nabla \cdot \underline{\underline{T}} = & \left(\frac{\partial T_{rr}}{\partial r} + \frac{1}{r} \frac{\partial T_{r\theta}}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial T_{r\varphi}}{\partial \varphi} + \frac{2T_{rr} - T_{\theta\theta} - T_{\varphi\varphi} + T_{r\theta} \cot \theta}{r} \right) \vec{e}_r \\ & + \left(\frac{\partial T_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial T_{\theta\theta}}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial T_{\theta\varphi}}{\partial \varphi} + \frac{3T_{r\theta} + (T_{\theta\theta} - T_{\varphi\varphi}) \cot \theta}{r} \right) \vec{e}_\theta \\ & + \left(\frac{\partial T_{r\varphi}}{\partial r} + \frac{1}{r} \frac{\partial T_{\theta\varphi}}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial T_{\varphi\varphi}}{\partial \varphi} + \frac{3T_{r\varphi} + 2T_{\theta\varphi} \cot \theta}{r} \right) \vec{e}_\varphi\end{aligned}$$